

The big picture

Guiding question(s)

What happens when electrons are transferred?

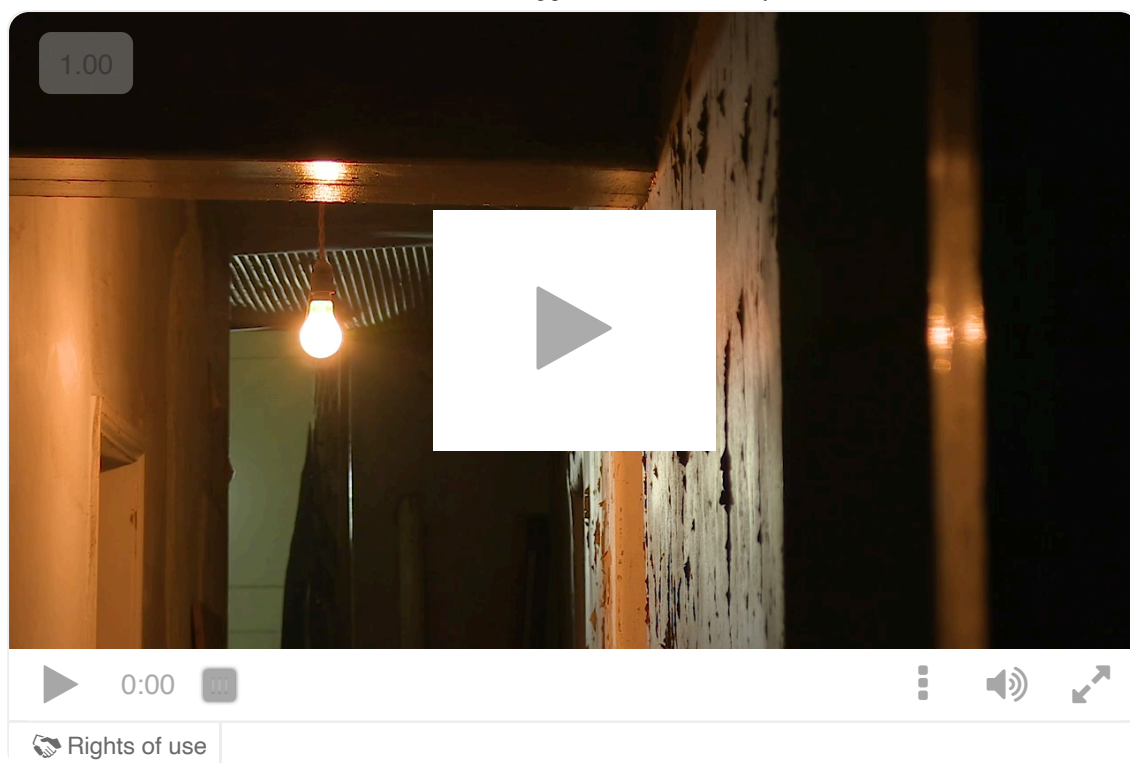
Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

You reach for your phone, but your battery is drained.



Figure 1. Dead battery symbol.

It's okay, you think you can recharge it. You get the plug, and as you place it in the socket...



Video 1. Power Outage.

👁 More information for video 1

The power supply goes out. If only you knew how to generate your own electricity.

What is electricity and what are batteries made from?

Electricity is the movement of electrons through a conducting material. You might recall that metals have electrons that can move freely (see subtopic S2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43111/>)), which makes them excellent conductors of electricity.

The Greek philosopher Thales of Miletus, around 600 BCE, observed that rubbing materials such as fur and amber, for example, would allow them to attract other objects. This phenomenon, known as static electricity, was a starting point for the research around electricity.

An interesting discovery of electricity use in antiquity is the Baghdad battery, dated 2000 years old. The small clay pot seems to have served as a wet battery, possibly used to generate a low voltage for medical procedures, such as electric shocks to treat patient pain.

During the mid-18th century, Benjamin Franklin did his famous experiment with a kite, hoping to show that lightning is a form of static electricity. Later, in 1800, an Italian doctor, Luigi Galvani, found that a frog's legs twitched if it touched two

different kinds of metals. Alessandro Volta, using this information, concluded that between two metal plates there exists a kind of electrical potential, which causes an electrical charge to travel through the legs of the frog. Volta invented the first modern battery known as a voltaic cell. Volta's battery included a stack of copper (Cu) and zinc (Zn) discs separated by cloth soaked in salty water. The units of electrical potential are called volts in honour of Volta.

A battery is a chemical energy storage in which a chemical reaction creates a flow of electrons that can power devices. Batteries have evolved a lot in recent years, using new materials in new formats. Today we have megawatt batteries, which store enough power to supply entire villages or islands, and tiny batteries like the ones used in electronic watches.

Concept

Electrons are transferred in oxidation-reduction reactions. When this happens spontaneously, energy is produced and can be either used or stored, as in batteries. Electricity can also be used to perform non-spontaneous reactions to obtain determined chemicals of great use to humans, like aluminium.

Prior learning

Before you study this subtopic make sure that you understand the following:

- How to describe electron arrangements for atoms according to their position on the periodic table (see subtopic S3.1  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>)).
- How to assign oxidation numbers (see section S3.1.6  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-states-id-44696/>)).
- Acids (see subtopic R3.1  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/>)), functional groups in organic compounds and unsaturated compounds (see subtopic S3.2  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/>)).
- Transition elements have delocalised d electrons (see section S2.3.3  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/transition-metals-hl-id-43565/>)).
- Change in Gibbs energy, ΔG , relates the energy that can be obtained from a chemical reaction to the change in enthalpy, ΔH , change in entropy, ΔS , and absolute temperature. At constant pressure, a change is spontaneous if the change in Gibbs energy, ΔG , is negative (see section R1.4.2–3 

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/>)).

Practical skills

Once you have completed this subtopic, apply your knowledge of electron transfer reactions by going to:

- Practical 9: Investigating factors affecting electrochemical cells
(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/investigating-factors-affecting-electrochemical-cells-id-46726/>)
- Practical 1.1.10a: Oxidation of alcohols
(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-of-alcohols-10a-id-46727/>)
- Practical 1.1.10b: Redox titration
(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/redox-titration-10b-id-46728/>).